

Internship Tracker - SRS

PRODUCT REQUIREMENT DOCUMENT

INTERNSHIP TRACKER

Internship Lifecycle Management System

Team Name	TECH MATRIX
Domain	Vibe Coding & Full Stack Development

A centralized digital platform for managing the complete internship lifecycle — from application to reward point generation.

Table of Contents

- | | |
|---------------------------------|-------------------------------------|
| 1. Executive Summary | 2. Problem Statement |
| 3. Objectives | 4. Stakeholders |
| 5. System Workflow | 6. User Journey |
| 7. Core Modules | 8. Smart Automation |
| 9. Functional Highlights | 10. Functional Requirements |
| 11. Non-Functional Requirements | 12. Technology Architecture & Stack |
| 13. Key Features | 14. Future Enhancements |
| 15. Expected Outcomes | 16. Project Impact |
| 17. Conclusion | |

1. Executive Summary

Internova is a centralized Internship Lifecycle Management System that digitizes and automates the complete internship process within higher educational institutions.

Students, mentors, HODs, Placement Cell members, and administrators collaborate through a structured approval workflow with complete visibility over internship activities.

Unlike systems that only handle registration and approval, Internova manages the entire lifecycle — application, approvals, progress tracking, report submission, verification, and reward point generation.

2. Problem Statement

Most colleges manage internships using paper forms, Google Forms, Excel sheets, email, WhatsApp groups, and manual document verification — leading to the pain points below.

Operational Pain Points Delayed approvals • Lost/duplicate documents • No internship visibility • No centralized tracking	Process Pain Points Manual report verification • Poor stakeholder communication • Difficult report generation
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Proposed Solution: Role-based access • Document management • Approval automation • Progress tracking • Analytics • Institutional reporting — all in one platform

3. Objectives

- ✓ Digitize the internship process & eliminate manual paperwork
- ✓ Improve transparency and automate approval workflows
- ✓ Track internship progress in real time
- ✓ Securely manage internship documents
- ✓ Simplify report verification & generate analytical reports
- ✓ Improve communication via notifications
- ✓ Automatically manage reward point approval

4. Stakeholders

Role	Responsibility
Student	Apply for internships, track progress, submit reports
Mentor	Verify internship details, monitor students, approve reports
HOD	Department-level approval and monitoring
Placement Cell	Final approval, analytics, reward verification
Administrator	Complete system management

5. System Workflow

End-to-end path of a single internship, from submission to completion:

Student → Submit Application → Mentor Review → HOD Approval → Placement Cell Approval → Approved → Internship Starts → Progress Tracking → Completed → Report Submission → Mentor Review → HOD Review → Placement Cell Review → Reward Points → Success

6. User Journey

Phase 1 — Internship Application

Uploads: Offer Letter • Resume • NOC (optional)

Mentor → **HOD** → **Placement Cell** → **Approved**

Phase 2 — Internship Tracking (auto-enabled on Start Date)

Student updates: Daily Work • Weekly Progress • Learning Outcomes • Hours Worked • Screenshots • Attendance

The mentor monitors progress throughout this phase.

Phase 3 — Internship Report (auto-enabled after End Date)

Uploads: Internship Report • Completion Certificate • Original Certificate • Attested Certificate • Presentation • Company Feedback

Mentor → **HOD** → **Placement Cell** → **Reward Points** → **Completed**

7. Core Modules

Authentication Login • Forgot Password • RBAC • Session Mgmt	Student Module Profile • Application • Tracker • Reports • Notifications • Reward Points
Mentor Module Student Monitoring • Approval • Progress Review • Report Verification	HOD Module Dept Dashboard • Approvals • Report Verification • Dept Analytics
Placement Cell Company Mgmt • Verification • Final Approval • Reports • Reward Approval	Administrator User Mgmt • Dept Mgmt • Role Mgmt • Analytics • System Config

8. Smart Automation

Feature	Before	After
Internship Tracker	Locked before start date	Auto-enabled on Start Date
Internship Report	Locked before internship end	Auto-enabled after completion

Notifications: Approval Updates • Start Reminder • Completion Reminder • Report Deadline • Reward Approval

9. Functional Highlights

Progress Tracking: Work Summary • Learning Progress • Tasks Completed • Weekly Reports

Document Management: Offer Letter • Completion Certificate • Internship Report • Presentation • Supporting Documents

Analytics Dashboard: Internship Status • Dept Analytics • Company Analytics • Completion Rate • Pending Approvals

10. Functional Requirements

Functional capabilities required to support the complete internship lifecycle:

10.1 Authentication & User Management

The system shall: Secure institutional login • Role-Based Access Control • Profile updates • Secure session management • Admin account management

10.2 Student Module

Profile: View/update profile • Academic info • Upload resume & documents

Application: Apply for internship • Enter company details • Upload offer letter • Save draft • Submit for approval • View status

Tracker (auto on Start Date): Daily/weekly progress • Tasks completed • Learning outcomes • Progress screenshots • Mentor remarks • Completion tracking

Report (auto after End Date): Submit report • Completion/original/attested certificates • Presentation (optional) • View approval status

Notifications: Approval updates • Start/completion reminders • Report deadline • Reward approval

10.3 Mentor Module

Mentors shall: View assigned students • Review applications • Verify documents • Approve/reject requests • Monitor progress • Review & approve reports • Add comments • View pending dashboard

10.4 Head of Department (HOD)

HOD shall: View dept applications • Review mentor recommendations • Approve/reject requests • Review reports • Monitor dept statistics • View analytics dashboard

10.5 Placement Cell Module

Placement Cell shall: Review approved requests • Verify company info • Final approval • Review reports • Approve reward eligibility • Generate institutional reports • View company/dept analytics

10.6 Administrator Module

Admin shall: Manage all user roles • Manage departments & companies • Configure academic years & workflow • Configure notifications • Manage document requirements • Monitor activity & audit logs • Generate reports

10.7 Approval Workflow

Student → **Mentor Review** → **HOD Approval** → **Placement Cell** → **Approved**

Report Submitted → **Mentor Verify** → **HOD Verify** → **Placement Verify** → **Reward Points** → **Completed**

10.8 Smart Automation

System shall auto: Enable tracker on start date • Lock tracker before start • Enable report after completion • Lock report before completion • Send approval reminders • Send deadline reminders • Flag overdue submissions

10.9 Document Management

Documents: Offer Letter • Internship Report • Completion/Original/Attested Certificates • Supporting Documents

Each supports: Preview • Download • Replace (pre-approval) • Version tracking (optional)

10.10 Dashboard & Analytics

Dashboards show: Internship statistics • Pending approvals • Progress • Completion rates • Dept/Company analytics • Monthly trends

11. Non-Functional Requirements

Category	Requirements
Performance	Dashboard loads ≤3s • concurrent access • fast upload/retrieval • efficient reports
Security	RBAC • password encryption • input validation • secure uploads • audit logs • HTTPS
Reliability	High availability • regular backups • graceful recovery • workflow data consistency
Scalability	Multi-department/year support • thousands of records • multi-institution ready
Usability	Responsive UI (desktop/tablet/mobile) • minimal steps • clear status indicators
Maintainability	Modular architecture • reusable components • documented code
Compatibility	Chrome, Edge, Firefox, Safari • desktop, tablet, mobile
Availability	24×7 uptime • automatic backups • minimal update downtime
Data Integrity	No duplicate applications • preserved approval history • secure document storage

12. Technology Architecture & Stack

Users → React App → Auth → Internship Mgmt → Approval Workflow → Documents → Notifications → Analytics → PostgreSQL DB

Layer	Technology
Frontend	React
Backend	Node.js
Language	TypeScript
Styling	Tailwind CSS
Database	SupaBase
Authentication	Auth.js
File Storage	Supabase Storage
Charts	Recharts
Deployment	Vercel
Version Control	GitHub

13. Key Features

★ Multi-Level Approval Workflow	★ Internship Lifecycle Tracking
★ Automated Progress Monitoring	★ Date-Based Feature Unlocking
★ Internship Report Verification	★ Document Management
★ Analytics Dashboard	★ Real-Time Notifications
★ Reward Point Automation	★ Role-Based Access Control

14. Future Enhancements

Roadmap: AI Resume Analysis • AI Internship Recommendation • Digital Signature • QR Certificate Verification • Mobile App • Company Portal • Attendance Integration • Calendar Integration • AI Chatbot • Predictive Analytics

15. Expected Outcomes

- ✓ Reduced paperwork & faster approvals
- ✓ Centralized, transparent internship records
- ✓ Better stakeholder communication
- ✓ Enhanced institutional reporting & secure documents
- ✓ Increased operational efficiency

16. Project Impact

 100% Digital Internship Workflow	 Reduces Approval Time by ~70%
 Real-Time Institutional Analytics	 Secure Role-Based Access
 Improved Student & Faculty Experience	 Complete Internship Lifecycle Management

17. Conclusion

Internova transforms the traditional internship process into a fully digital, transparent, and automated workflow — from application and approvals to progress tracking, report verification, and reward point generation.

Its scalable architecture and user-centric design make it a sustainable solution for modern college management.

System Design and Architecture

INTERNSHIP TRACKER

System & Software Architecture Document

1. High-Level System Architecture

Users

Student • Mentor • HOD • Placement Officer • Administrator



React Web Application (Frontend)

Authentication Pages • Dashboard • Internship Module • Company Module • Document Module • Reports • Analytics • Notifications



Node.js Route Handlers / Server Actions

Authentication • Internship Service • Approval Service • Company Service • Document Service • Notification Service • Report Service



Business Logic Layer

RBAC Authorization • Workflow Engine • Validation • File Processing • Report Generator



Prisma ORM

Type-safe query builder & migrations



Supabase Database

Users • Roles • Students • Mentors • Departments • Companies • Internship Applications • Documents • Reports • Notifications • Audit Logs

2. Software Design Architecture (Layered Architecture)

Presentation Layer

React App Router • React Components • Tailwind CSS • Shadcn UI



Application Layer

Route Handlers • Server Actions • Authentication • Validation



Business Layer

Internship Workflow • Company Management • Document Verification • Approval Process • Analytics • Notification Engine



Data Access Layer

Repository Pattern • Database Queries



Database Layer

SupaBase

3. Module Architecture

All modules are organized around the central Internship Portal:

▸ Authentication Module	▸ Student Module
▸ Mentor Module	▸ HOD Module
▸ Placement Cell Module	▸ Admin Module
▸ Internship Module	▸ Company Module
▸ Document Module	▸ Notification Module
▸ Reports Module	▸ Analytics Module

4. Approval Workflow Architecture

Student → Submit Internship → Mentor Review → Approve/Reject
→ HOD Review → Approve/Reject → Placement Cell Review →
Approve/Reject → **Approved**

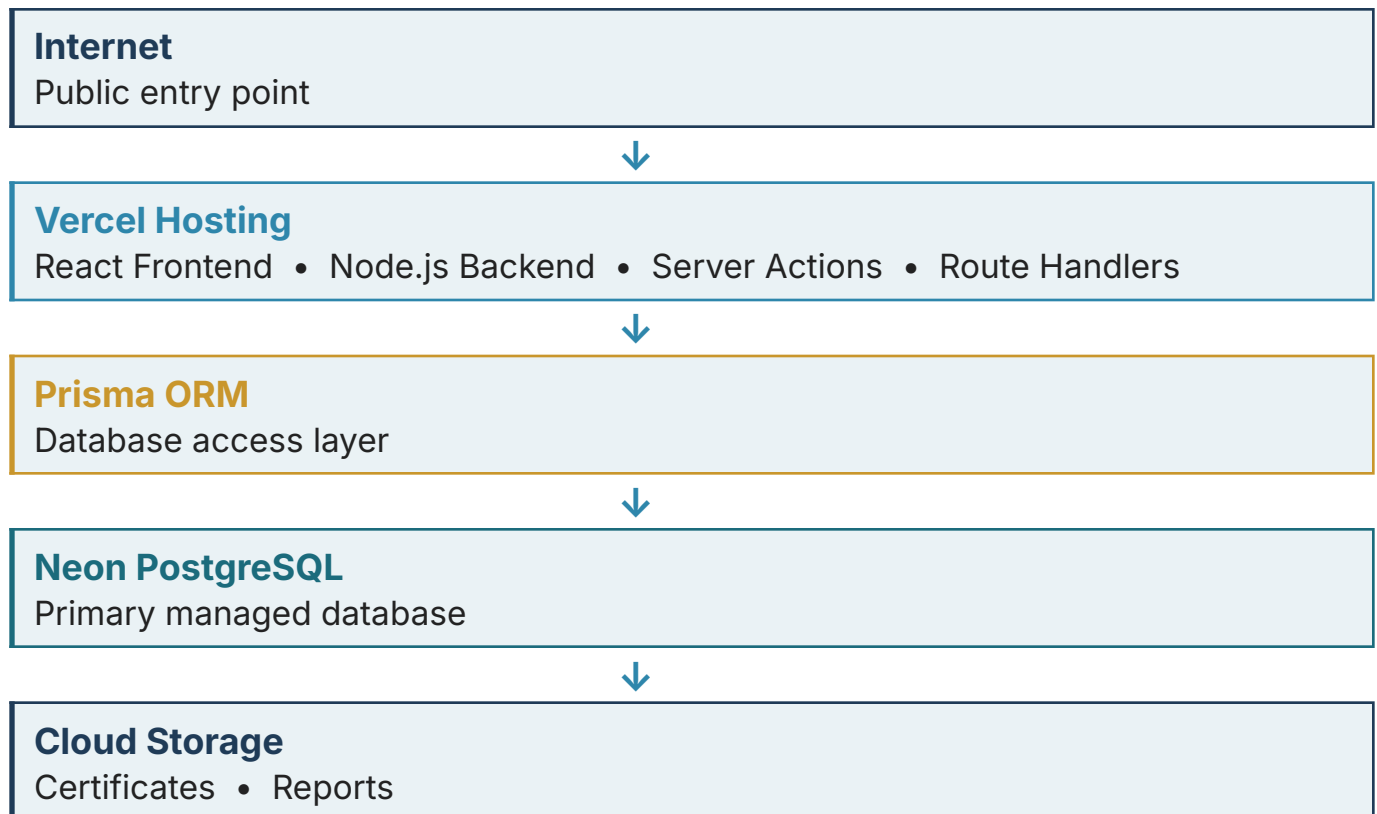
Starts Internship → Weekly Reports → Final Report → Completion
Certificate → **Completed**

5. Database Architecture

Core entity relationships across the schema:

Parent Entity	Related / Child Entities
Users	Roles, Students, Mentors, HOD, Placement Members
Students	Internship Applications, Documents, Notifications, Reports
Internship Applications	Company, Mentor Approval, HOD Approval, Placement Approval, Status

6. Deployment Architecture



7. Component Architecture

Components grouped by tier:

Frontend	Backend	Database
Login	Auth Service	Authentication
Dashboard	Internship Service	SupaBase
Forms	Approval Service	
Charts	Document Service	
Tables	Analytics Service	
Notifications	Report Service	